

Diluted Regional Citrate Anticoagulation for Continuous Renal Replacement Therapy in Pediatric Patients: A Systematic Review and Suggested Practice Points

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Keywords

Continuous renal replacement therapy · Children · Citrate · Anticoagulation · Acute kidney injury · Continuous venovenous hemofiltration

Abstract

Introduction: Continuous renal replacement therapy (CRRT) is increasingly used in critical pediatric patients with acute kidney injury (AKI). The choice of anticoagulant is vital to minimize circuit clotting and bleeding complications. Regional citrate anticoagulation (RCA) is preferred for its safety profile, particularly in critically ill pediatric patients who are susceptible to bleeding. **Methods:** A comprehensive literature search was conducted using PubMed, Google Scholar, and Cochrane databases following PRISMA guidelines. Keywords included "diluted citrate," "regional citrate anticoagulation," "continuous renal replacement therapy," "pediatrics," and "adverse effects." Studies were included if they involved

neonates and pediatric patients, reported citrate concentration, and safety and efficacy outcomes of RCA in CRRT. Data were extracted on study characteristics, citrate concentration, circuit lifespan, metabolic and electrolyte disturbances, and other adverse effects. **Results:** A total of 16 studies met the inclusion criteria. RCA was associated with fewer clotting events and a longer median circuit life compared to heparin. However, complications such as metabolic alkalosis, hypocalcemia, and hypernatremia were noted. In our single-center experience, dilute citrate anticoagulation was used in 16 pediatric patients undergoing CRRT, showing promising results with reduced clotting and prolonged circuit life. The modified pediatric citrate protocol presented aims to address complications by using a diluted citrate solution. **Conclusions:** RCA is effective in prolonging circuit life and reducing clotting in pediatric CRRT. The

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modified pediatric citrate protocol presents a safer alternative by reducing the risk of metabolic and electrolyte disturbances. Ongoing monitoring of calcium and electrolyte levels is essential to mitigate potential complications. This protocol may standardize RCA use in pediatric CRRT, improving safety and outcomes for critically ill children with AKI.

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Introduction

The use of continuous renal replacement therapy (CRRT) in pediatric patients has steadily risen in recent years, and the choice of anticoagulant is a crucial consideration to address the risk of circuit clotting [1]. Regional citrate anticoagulation (RCA) is a well-established anticoagulation strategy used in CRRT and is associated with fewer incidences of bleeding [2, 3]. This is particularly important in critically ill pediatric patients who are already at risk for bleeding complications.

Citrate has a favorable pharmacokinetic profile in pediatric patients [4]. It is rapidly metabolized in the liver and its metabolites are excreted in the urine. Citrate can be easily given within the CRRT circuit pre-blood pump. This is important for critically ill pediatric patients who may require multiple lines and invasive procedures [5, 6].

Citrate preparations and solutions vary greatly in their concentrations, and ergo numerous RCA protocols have been created, lending to the heterogeneous guidelines and difficulty of comparing different trials and centers. Most RCA protocols for children typically use concentrated citrate, and not many protocols have been described using dilute citrate. This review aims to discuss citrate use as an anticoagulation strategy in pediatric CRRT, with a focus on safety, efficiency, patient outcome, and adverse events, as well as the availability and feasibility of dilute citrate solution in critical children with acute kidney injury (AKI). The current review also includes a protocol for the use of diluted RCA in children, with departmental audit of the use in children.

Materials and Methods

Search Strategy

A comprehensive search of PubMed, Google Scholar, and Cochrane databases was conducted by two independent reviewers, in accordance with Preferred Re-

porting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines (online suppl. Table 1; for all online suppl. material, see <https://doi.org/10.1159/000545674>). We searched using the following keywords: “diluted citrate,” “regional citrate anticoagulation,” “continuous renal replacement therapy,” “pediatrics,” “hemodialysis,” and “adverse effects.” We also searched the reference list of the retrieved studies and articles to include additional publications. The last search was performed on August 31, 2022 and only English written manuscripts were considered.

Study Selection

Two independent reviewers assessed the literature results separately and any conflicts and disagreements were discussed with a third reviewer.

• *Inclusion criteria:* The inclusion criteria were as follows:

- Neonates and pediatric age group patients.
- Regional citrate was used as a method of anticoagulation in renal replacement therapy and its concentration was reported.
- Safety-related outcomes such as metabolic and electrolyte disturbances were reported.
- Efficacy in terms of median circuit lifespan or filter span was reported.

There was no restriction on time duration or geographic location. Cross-sectional studies, prospective studies, and retrospective studies were primarily included in the review to assess the use of diluted regional citrate as an anticoagulation method on pediatric patients undergoing CRRT.

• *Exclusion criteria:* The exclusion criteria were as follows:

- Studies which did not report concentration of RCA administered or used other methods of anticoagulation in RRT.
- Safety-related outcomes and efficacy were not studied.
- Reviews articles, systematic reviews, comments, and letters and case reports.

Data Extraction and Quality Assessment

Data extracted from the included studies consisted of the following: first author, title, year of publication, sample size, gender, mean age, country of study, study design, concentration of citrate used, other methods of anticoagulation used, mean circuit lifespan, filter lifespan, total CRRT duration, metabolic and electrolyte disturbances, other adverse effects, citrate accumulation, patient mortality, and length of stay in

neonatal intensive care unit/pediatric intensive care unit. For statistical analysis, R software (version 4.1.4), package metafor was used. The data were expressed as mean difference with 95% CI in case of continuous data and odds ratio (OR) with 95% CI in case of categorical data.

Results

A total of 812 studies were found by our initial search on the various databases. Of these, 16 studies matched our inclusion criteria and were used for this review. A PRISMA flow diagram is shown in Figure 1 to illustrate the study selection process.

Basis of Regional Citrate Anticoagulation in CRRT

The preferred method for managing critically ill children with AKI and fluid overload is CRRT [5, 6]. Due to the small size and body surface area, the extracorporeal circuit volume, the requirement for circuit blood priming, and equipment adaption, administration of CRRT in children is considered to be challenging. Anticoagulation is commonly used to prevent clotting and extend the lifespan of CRRT circuits, with unfractionated heparin and sodium citrate considered the most popular options. In recent years, RCA has emerged as the gold standard for treating adult patients with AKI [7, 8]. In comparison to heparin anticoagulation, RCA has been found to lower the risk of bleeding and increase circuit lifespan in adult studies [9–15]. Citrate as an anticoagulant is preferred due to its ability to chelate ionized calcium, which, in turn, leads to inhibition of the coagulation cascade. A number of cofactors are impacted by the depletion of functional calcium, preventing the propagation of coagulation cascade and ultimately inhibition of thrombin formation [8]. Calcium must be present to begin the clot-forming process, hence reducing calcium can lengthen the life of the hemofilter [9]. Citrate's molecular weight of 191 Da lends itself as a small molecule in comparison to unfractionated heparin, which ranges from 3,000 to 30,000 Da, and low molecular weight heparin, which ranges from 2,000 to 9,000 Da. The compact size of citrate attributes to its high diffusion rate across the membrane during dialysis, resulting in it predominantly being lost in dialysate effluent. However, since most of the calcium-citrate complex is filtered during dialysis, a calcium infusion is necessary to replenish what is lost either through a separate venous line or the through a dialysis venous line [3]. Moreover, the patient's liver, kidney, and skeletal muscle metabolize the remaining calcium-citrate com-

plexes into bicarbonate [10]. In a recent systematic review comparing RCA and heparin anticoagulation in CRRT in pediatric patients (11 studies, cumulative 1,706 CRRT sessions [62% systemic anticoagulation and 38% RCA]), RCA provided a longer circuit lifespan with a significantly decreased clotting rate and was superior to heparin, yet gives rise to metabolic alkalosis and alterations in calcium metabolism [16].

Pediatric Studies on RCA Compared to Heparin

We found 16 studies that administered RCA in pediatric CRRT (in children <18 years of age), as shown in Table 1. Of the studies evaluating circuit life and clotting of filters as a treatment outcomes, it was reported that RCA had a longer median circuit lifetime as compared to heparin. Sik et al. [9] found 11.36% of patients experienced clotting of the CRRT circuit with RCA, while 15/26 (26.31%) experienced clotting of the CRRT circuit with heparin (OR: 0.36; 95% CI: 0.12, 1.08). The odds of circuit clotting was 3.5 (95% CI: 1.09, 11.68) with heparin as compared to RCA counterparts. No difference was noted in mortality rate, which was 6/19 (31.58%) in the RCA group and 8/29 (30.77%) in the heparin group [9]. Zaoral et al. [17] found that RCA gave rise to significantly longer circuit lifetime (median time 41.0 h; maximum duration 72 h) compared to heparin (median time 36.0 h; maximum duration 56 h), with age, weight, and blood flow rate (BFR) shown to be influential parameters, $p = 0.0001$. The group also reflected that strategies that used RCA were associated with fewer transfusion rates. Rico et al. [18] reported that hemofilter survival was significantly higher in the RCA cohort: 72 h after RCA and 18 h after heparin ($p < 0.0001$). Although circuits treated with RCA filters had lower BFRs (43.9% of RCA filters had a BFR <100 mL/min as opposed to 17.3% of the heparin filters), RCA was associated with a longer circuit lifetime. Raymakers-Janssen et al. [19] reported a median circuit survival of 45.2 h with RCA and 21 h with heparin ($p < 0.01$). In the subgroup analysis, clotting caused hemofilter failure in 42% of heparin filters compared to 17.1% of the RCA filters ($p < 0.001$). Fernández et al. [15] found that the median circuit survival was 48 h with RCA and 31.0 h with heparin ($p = 0.028$). Total of 76.4% of heparin filters clotted, compared to 18.8% of RCA filters ($p < 0.01$). Soltysiak et al. [14] also reported that circuit life was significantly higher in the RCA cohort: 58.04 h compared to 37.64 h ($p = 0.03$). Circuit clotting was seen in 11.63% of patients in the RCA group, compared to 34.15% in the heparin group. The Kaplan-Meier analysis of circuit lifetime probability found higher circuit survival for RCA CRRT system as compared

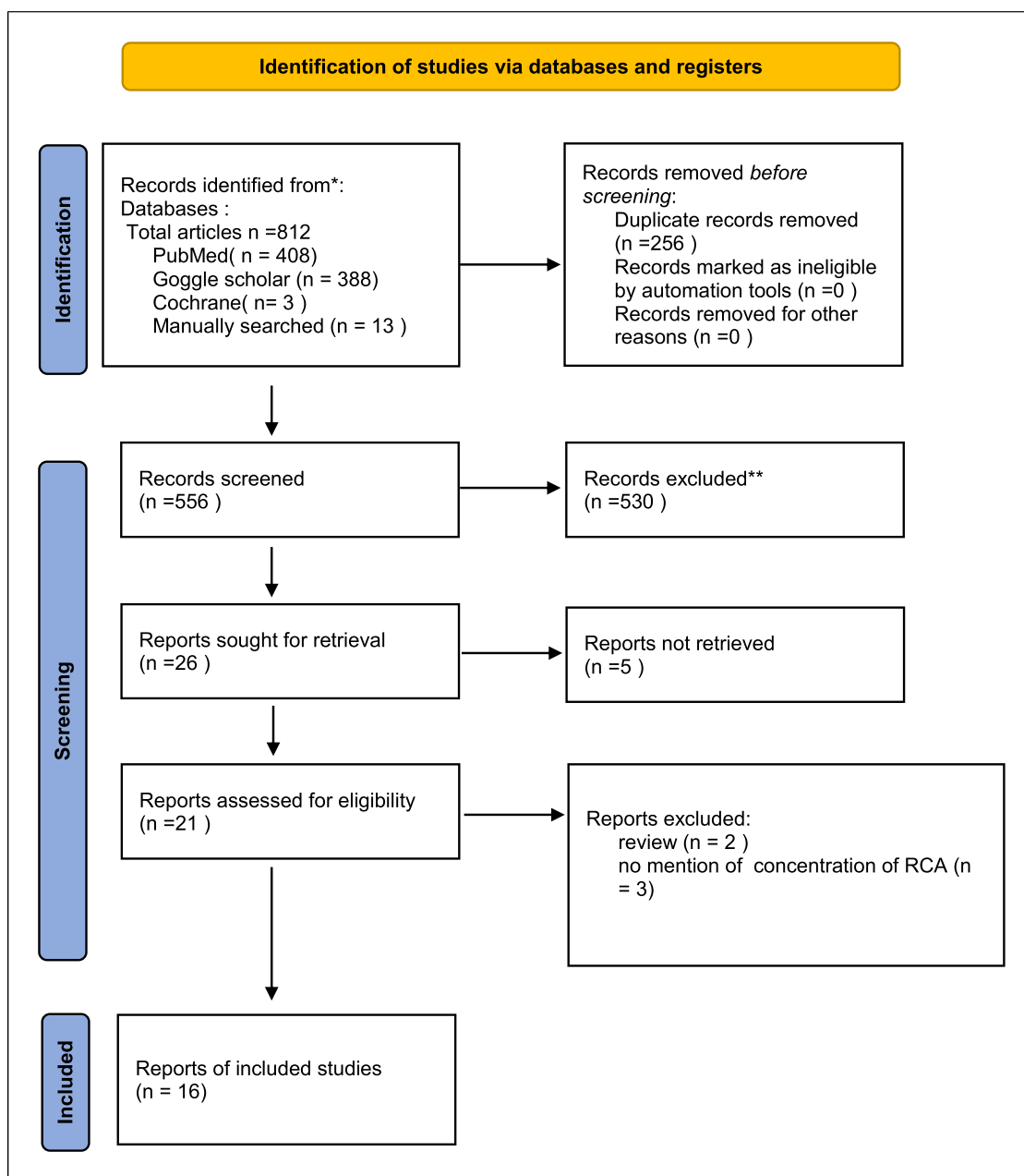


Fig. 1. Criteria of the reviewed studies.

to the Heparin Anticoagulation (HA) CRRT system (41.86% of RCA CRRT circuits functioned longer than 72 h vs. only 21.95% of those used in HA CRRT).

Two studies found that heparin was associated with a lesser risk of clotting. Cortina et al. [21] reported that heparin use was associated with the longest mean circuit life of 29.3 h, followed by RCA at 23.9 h, heparin/protamine at 17.7 h, and no anticoagulant at 8.5 h ($p < 0.001$). Kakajiwala et al. [20] conducted a retrospective cohort study of 26

patients in which heparin administration was associated with a lower risk of clotting ($p = 0.006$). Of the 28 circuits that underwent scheduled change, 2/28 occurred in the heparin group vs. 25/28 in the RCA group (RR: 0.08, 95% CI: 0.02, 0.31). A total of 1/28 circuit did not utilize any anticoagulation. Fifty-one circuits underwent unplanned change due to filter clotting, 26 (51.0%) in the heparin group and 20 (39.2%) in the RCA group (OR: 0.62, 95% CI: 0.28, 1.36). A total of 5/51 circuits had no anticoagulation.

Table 1. Characteristics of the 16 selected studies

Study/study period	Country	Study design	Eligible patients, <i>n</i>	Male sex, %	Age	Body weight, kg	Total CRRT duration, h	Regional citrate concentration, mmol/L
Miklaszewska et al. [11]/January 2009 to June 2016	Poland	Comparative retrospective observational	40	58	Mean: 116.7 months	Mean: 34.8	7,637	18
Liet et al. [12]/December 2011 to December 2012	France	Retrospective observational	7	–	Mean: 9.85 years	Mean: 38.14	330	18
Persic et al. [13]/2011 to 2016	Slovenia	Retrospective observational	10	70	Mean: 2.6 months	Mean: 4.6	–	18
Soltysiak et al. [14]/January 2012 to December 2012	Poland	Comparative retrospective observational	30 (16 in RCA group and 14 in HA group)	50	Mean: 19.4 months	Mean: 8.69	–	18
Fernández et al. [15]/January 2011 to February 2013	Spain	Comparative retrospective observational	36 (12 in RCA group and 24 in HA group)	–	Median RCA: 36 months Median heparin: 33 months	Median RCA: 36.0 Median heparin: 33.0	–	12, 18
Buccione et al. [16]/2012 to 2017	Italy	Retrospective observational	28	46.4	Median: 2 years	Median: 13.0	–	18
Kakajiwal et al. [20]/January 2013 to December 2014	USA	Comparative retrospective observational	26	58.2	Median: 11.8 years	Body surface area: 0.96	–	74.8
Rico et al. [18]/January 2008 to July 2013	Columbia	Comparative retrospective observational	32 (RCA 17; HA group 15)	37.5	Mean RCA: 8.8 years Mean heparin: 9.35 years	Mean RCA: 29.14 Mean heparin: 33.6	3,442	74.8
Raymakers-Janssen et al. [19]/November 2011 to October 2016	Netherlands	Comparative prospective observational	20 (RCA group: 14 and HA group: 6)	55	Median RCA: 17 months Median heparin: 13 months	Median RCA: 9.8 Median heparin: 9.0	4,530	74.8
Cortina et al. [21]/2007 to 2014	Australia	Comparative retrospective observational	161	52.8	Median: 5.1 years	Median: 20	22,190	74.8
Rodriguez et al. [22]/30-month timeframe	USA	Retrospective observational	51	39	Median: 3.5 years	Median: 13.8	861 days	74.8
Elhanan et al. [23]/12-month timeframe	Canada	Retrospective observational	9	66.6	Mean: 8.8 years	Mean: 30.4	–	74.8

Table 1 (continued)

Study/study period	Country	Study design	Eligible patients, <i>n</i>	Male sex, %	Age	Body weight, kg	Total CRRT duration, h	Regional citrate concentration, mmol/L
Bunchman et al. [24]/November 2000 to March 2001	USA	Prospective observational	9	–	–	Mean: 31.6	–	74.8
Zaoral et al. [17]/January 2009 to December 2014	Czech Republic	Comparative crossover trial	63	54	Mean: 89.24 months	Mean: 30.37	9,381	136
Sik et al. [9]/August 2016 to August 2018	Turkey	Comparative retrospective observational	45 (19 in the RCA group and 26 in the HA group)	67	Median RCA: 77.5 Median heparin: 66.5	Median RCA: 15 Median heparin: 13	5,082	136
Huang et al. [25]/September 2015 to January 2021	China	Retrospective observational	23	61	Mean: 10.1 days	Mean: 3.0	–	136

CRRT, continuous renal replacement therapy; HA, Heparin Anticoagulation.

One study found no significant difference between cohorts in filter survival time. Miklaszewska et al. [11] conducted a retrospective study in which 29/66 (43.9%) CRRT filters were clotted in the RCA cohort, while 25/84 (29.8%) were clotted in the heparin cohort (OR: 1.85; 95% CI: 0.94, 3.6). Moreover, this study showed that clotting in the CRRT circuit was 2-fold higher in patients treated with the smaller HF20 hemofilter as compared to the larger ST60 hemofilter, and more than five times higher as compared to large ST100 filters regardless of the anticoagulation used. Thus, the study shows that filter clotting was higher with smaller filter sizes irrespective of the type of anticoagulation used. The mortality rate was 3/8 (38%) in the RCA cohort and 21/32 (66%) in the heparin cohort (RR: 0.57; 95% CI: 0.23, 1.45) without any significant difference.

A total of 6 of 16 studies had used 0.5% trisodium citrate (TSC), which constitutes 18 mmol/L of citrate in the anticoagulation solution. Three studies used 136 mmol/L of citrate, while 7 used the more commonly administered ACD-A solution, which comprises 74.8 mmol/L of citrate (Table 1).

Concentrated Citrate Solution

Concentrated citrate protocols used in the children and adult studies employed solutions with different citrate concentrations [26–30]. Undiluted 4% TSC solution

(sodium 408 mmol/L; TSC 136 mmol/L) was utilized by Mehta et al. [31] in 1990 for CRRT. Due to this solution's hypertonicity in terms of sodium and alkali concentrations, hyponatremia and metabolic alkalosis were common complications associated with the usage of 4% TSC as an anticoagulant. To compensate for the high concentrations of sodium and alkali in the anticoagulant, Mehta et al. [31] used a dialysate with hypotonic concentrations of the same.

ACD-A (sodium 224 mmol/L; TSC 75 mmol/L; citric acid 38 mmol/L; and dextrose 139 mmol/L) was later adapted for use as a regional anticoagulant in CRRT from the original usage in blood banks and apheresis procedures. ACD-A solutions with 112.9 mmol/L citrate (3.22%) and 224.4 mmol/L sodium were utilized in CVVHDF in a 2004 study by Cointault et al. [32]. ACD-A solutions are less prone to result in citrate toxicity and other issues than TSC because they have lesser concentration of TSC but have a high concentration of glucose.

Dilute Citrate Solutions

In 2006, Tolwani's protocol contrasted the application of a 0.67% TSC solution (140 mEq/L sodium and 23 mmol/L citrate) with a 0.5% TSC solution (140 mEq/L sodium and 18 mmol/L citrate) [33]. There were no significant variations in anticoagulant effectiveness; however, metabolic alkalosis and hyponatremia were

Table 2. Intensive care unit length of stay, outcome measures, results, and metabolic complications of the 16 studies

Study	Regional citrate concentration, mmol/L	Length of stay in PICU	Outcome measures	Results	Complications with RCA	Remarks
Miklaszewska et al. [11] (2017)	18	Mean: 19.0 days Median: 16.5 days	Filter size, filter pressure, and anticoagulant choice to determine circuit survival	No significant difference was noted in filter survival time between groups using RCA and heparin anticoagulation. 43.9% of RCA filters clotted and 29.8% of heparin filters clotted. Smaller filter size gave rise to shorter circuit life	–	Patients experienced a nonsignificantly increased risk in mortality with heparin (66%) compared to RCA (38%)
Liet et al. [12] (2014)	18	–	Hemofilter run time, percentage decrease in BUN and creatinine levels in 1st 24 h of CRRT, metabolic complications	Median filter lifetime of 24.0 h, median % decrease in BUN and creatinine in 1st 24 h of CRRT is 42.7% and 49.4%	Metabolic alkalosis	No incidence of citrate accumulation or any other metabolic complications with usage of integrated citrate software
Persic et al. [13] (2020)	18	–	Circuit lifetime, metabolic complications, and citrate accumulation	70% of patients saw citrate accumulation. Increased total/ionized calcium was seen in 25% of circuits	Metabolic acidosis (29%) Hyponatremia (27%) Hypernatremia (4%) Hypophosphatemia (65%) Hypomagnesemia (27%)	Logistic regression and receiver-operating characteristic curve calculated a citrate loading cutoff of 1.7 mmol/h/bw to predict citrate accumulation
Soltysiak et al. [14] (2014)	18	–	Circuit lifetime and metabolic complications	Circuit longevity was significantly higher for the RCA cohort (58.04 h) compared to the heparin cohort (37.64 h) ($p = 0.03$)	Metabolic alkalosis (25%) Metabolic acidosis (43.75%) Hypocalcemia (43.75%) Hypercalcemia (43.75%) Hypokalemia (60%) Hypernatremia (18.75%) Hyponatremia (18.75%)	Children receiving RCA-CRRT had a survival rate of 37.5%, whereas those receiving heparin had a survival rate of 14.3%

Table 2 (continued)

Study	Regional citrate concentration, mmol/L	Length of stay in PICU	Outcome measures	Results	Complications with RCA	Remarks
Fernández et al. [15] (2014)	18	Median RCA: 17 days Median heparin: 22 days	Circuit lifetime and metabolic complications	Average circuit survival was 48 h in the RCA group and 31 h in the heparin group ($p = 0.028$)	Metabolic alkalosis (66.6%) Hypochloremia (45%) Hypomagnesemia (27.3%)	Clotting was the most common reason for circuit changes in heparin filters (76.4%) compared to the citrate group (18.8%) ($p < 0.01$)
Buccione et al. [16] (2022)	18	–	Circuit lifetime and short- and long-term outcomes	18.2% of RCA filter clotted, 60.6% of heparin filters clotted, and 72.2% of filters with no anticoagulant clotted	–	Multivariate Cox analysis determined RCA as independent factor for premature clotting of the circuit ($p = 0.008$)
Kakajiwal et al. [20] (2017)	74.8	–	Patient characteristics and circuit-related variables (blood flow, dialysate flow, filter replacement fluids, and anticoagulant use) on circuit lifetime	Heparin was associated with a lower risk of clotting vs. RCA ($p = 0.006$)	–	–
Rico et al. [18] (2017)	74.8	Median RCA: 8.5 days Median heparin: 17.5 days	Circuit lifetime and metabolic complications	The longevity of the hemofilters in the citrate group was greater than in the heparin group, with a mean difference of 54 h, using Cox's proportional risk analysis ($p < 0.0001$)	Metabolic alkalosis (86.6%) Hypocalcemia (60%) Hypomagnesemia (46.6%) Hypercalcemia (13.3%)	Despite a lower blood flow, citrate was associated with a longer hemofilter lifetime. (43.9 vs. 17.53%)
Raymakers-Janssen et al. [19] (2017)	74.8	Median RCA: 15 days Median heparin: 26 days	Circuit lifetime, metabolic complications, and number of red blood cell transfusions	Average circuit survival time with RCA (45.2 h) was higher than with heparin (21 h) ($p < 0.01$)	Metabolic alkalosis (3%)	Clotting caused hemofilter failure in 42.0% of heparin-utilized filters compared to 17.1% of citrate-utilized filters ($p < 0.001$)
Cortina et al. [21] (2020)	74.8	Median: 11.0 days	Patient characteristics and anticoagulant on circuit survival	Heparin anticoagulation was associated with longer circuit life (29.3 h) followed by RCA (23.8 h) ($p < 0.001$)	–	Patients with initial admission of hematological malignancy experienced the highest mean circuit lifespan at 41.9 h ($p < 0.001$)

Table 2 (continued)

Study	Regional citrate concentration, mmol/L	Length of stay in PICU	Outcome measures	Results	Complications with RCA	Remarks
Rodriguez et al. [22] (2017)	74.8	Median: 27 days	Circuit lifetime, complications, and citrate accumulation	Median filter life was 66 h with RCA. 15% of filters were lost to clotting	Metabolic acidosis (17.6%) Metabolic alkalosis (17.6%) Hypocalcemia (6%) Hypercalcemia (69.4%)	70% of patients experienced citrate accumulation. Hospital mortality was 57%
Elhanan et al. [23] (2004)	74.8	–	Circuit lifetime and metabolic complications	Mean circuit life was found to be 55.6 h	Mild metabolic alkalosis (66.6%)	Mortality rate was 77%
Bunchman et al. [24] (2002)	74.8	–	Circuit patency and complications	The circuit life was determined to be 71.3 h with RCA	Metabolic alkalosis (100%) Hyponatremia (11%) Hypernatremia (11%) No bleeding occurrences	–
Zaoral et al. [17] (2016)	136	–	Anticoagulation choice and patient characteristics associated with circuit survival	The median circuit lifetime was significantly higher for RCA (41.0 h) than heparin (36.0 h)	Hypocalcemia (6.78%) Hypercalcemia (3.4%) Hypernatremia (6.77%) Metabolic acidosis (11.11%) Metabolic alkalosis (5.1%)	Kaplan-Meier survival curve of the median filter lifetime showed a higher survival time for RCA vs. heparin groups Transfusion rates varied significantly between anticoagulation strategies with a lesser requirement with RCA ($p = 0.03$)
Sik et al. [9] (2020)	136	Median RCA: 15 days Median heparin: 20 days	Circuit lifetime, transfusion requirements, and metabolic complications	In comparison to heparin, median circuit lifespan for RCA was significantly longer. When heparin was used as an anticoagulant, the risk of hemofilter clotting was 2.7 times higher than citrate, according to Cox's proportional risk analysis ($p < 0.001$)	Metabolic alkalosis (7.01%) Hypocalcemia (12.28%) Hypernatremia (14.03%)	A longer hemofilter life was found with the RCA group as compared to systemic anticoagulation ($p < 0.025$)

Table 2 (continued)

Study	Regional citrate concentration, mmol/L	Length of stay in PICU	Outcome measures	Results	Complications with RCA	Remarks
Huang et al. [25] (2023)	136	–	Circuit lifetime, metabolic complications, and citrate accumulation	Mean filter life was reported as 31.54 h with RCA	Metabolic alkalosis (72.5%) Metabolic acidosis (7.8%)	The risk of citrate accumulation in neonates given RCA was not greater than that of children or adults and was not related to age or metabolism

PICU, pediatric intensive care unit; RCA, regional citrate anticoagulation; bw, body weight.

more common in the group of 0.67% citrate. This investigation confirmed that using concentrated citrate solutions carries a comparatively higher risk of hypernatremia and metabolic alkalosis. Usage of dialysate or replacement fluids with abnormally low sodium and bicarbonate concentrations is also necessary for many of these concentrated solutions.

Discussion

There are various complications associated with the use of citrate as an anticoagulant in pediatric CRRT. Among the studies, the most common were metabolic abnormalities: metabolic alkalosis/acidosis followed by hypocalcemia, hypernatremia, and hypomagnesemia.

Metabolic Abnormalities

Each mole of citrate is metabolized to 3 moles of bicarbonate in the liver, which can potentially lead to metabolic alkalosis. As a result, metabolic alkalosis is the most frequent citrate-related metabolic complication [34]. Citrate may also cause metabolic acidosis, which is especially problematic in critically ill pediatric patients who are already at risk for acid-base imbalances. Metabolic acidosis often occurs as a result of inappropriate buffer supply to adjust for pH imbalances. If there is a disparity between the amount of citrate/bicarbonate administered and the quantity of citrate/bicarbonate removed in the effluent fluid, metabolic acidosis ensues [35]. Another possible mechanism for metabolic acidosis is impaired citrate metabolism, which may arise from sepsis, shock, or organ failure that results in lactic

acid production [36]. Citrate accumulation in the blood and subsequent metabolic acidosis and hypocalcemia has been linked to citrate toxicity in individuals with liver function impairment (low levels of systemic ionized calcium, as total calcium may be normal) [34–36].

Citrate Lock

Citrate accumulation or lock is seen when the administration of citrate exceeds the body's capacity to metabolize it, e.g., in liver failure and circulatory shock states. It is most commonly suspected with the increasing total/ionized calcium ratio. This ratio reflects an increase in serum anion-bound calcium. Another suggestion of citrate accumulation is an increase in the need for calcium supplementation due to continuous chelation of calcium with citrate forming a complex which is not effectively metabolized. The patients with liver failure and lactic acidosis due to inability to metabolize citrate also develop high anion gap metabolic acidosis [18, 23].

In relation to children, citrate accumulation is commonly seen in neonatal and infant population. Liver and skeletal muscles often break down citrate rapidly. The liver is very immature in newborns and infants, and critically ill children frequently exhibit hepatic dysfunction and hypoperfusion brought on by shock. Additionally, only 10–30% of the citrate is eliminated by CRRT, in contrast to adult patient in which elimination is approximately 40–60% [13, 18]. Further, due to technological constraints of adult CRRT devices, pediatric patients require different considerations and adjustments in their CRRT settings. For instance, citrate dose, which is dependent on BFR, is extraordinarily high (per kg of body weight) in pediatric patients, as compared to adults. This difference is more pronounced in smaller children [13, 18].

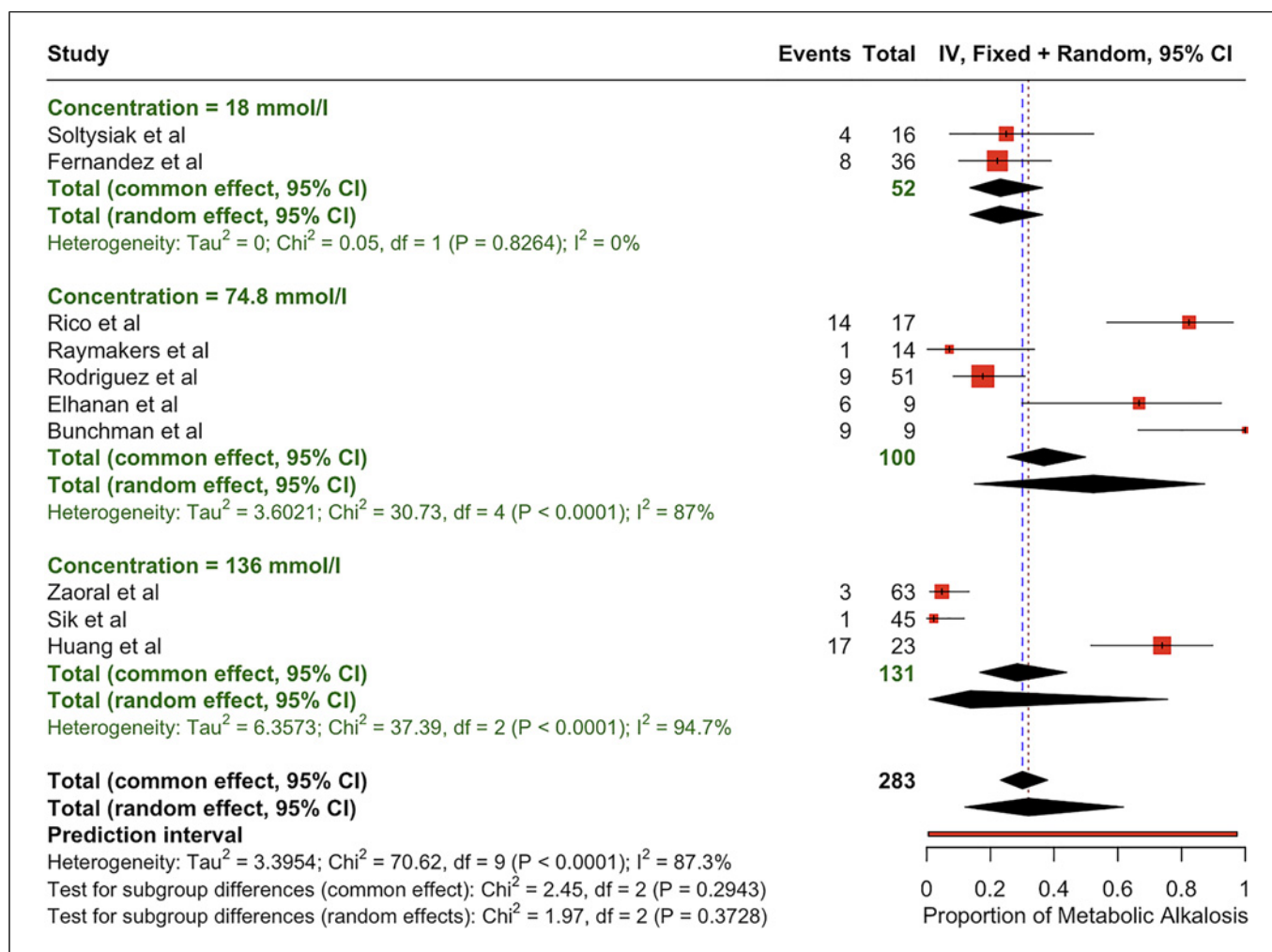


Fig. 2. Adverse events: metabolic alkalosis between different RCA concentrations.

Frequent citrate accumulation has been observed in prior studies (30% of patients in Rico et al. [18] had $\text{tCa}:\text{iCa} > 2.4$), particularly in patients with liver failure (70% in Rodriguez et al. [22] had $\text{tCa}:\text{iCa} > 2.5$ for more than 48 h in 51 pediatric liver failure patients). However, Huang et al. [25] found no difference in the risk of citrate accumulation in neonates relative to children or adults and reported no link between accumulation and gestational age, contradicting the results of Persic et al. [13]. Regardless, careful monitoring of electrolyte levels is necessary to prevent this complication.

Hypocalcemia

Citrate use can also cause hypocalcemia, which can result in dangerous conditions like hypotension and arrhythmias, including ventricular tachycardia due to irregular contractility of the heart [1, 26–28]. Other conditions, such as tetany (muscle spasms) and muscle

weakness, can also occur. This is due to the chelation of calcium ions by citrate, which is necessary for the formation of blood clots, and exacerbated by inappropriate calcium replacement. It is associated with dialytic removal of calcium citrate and the usage of calcium free dialysate solution. In the setting of impaired citrate metabolism, systemic ionized calcium levels decrease due to chelation of citrate with calcium, resulting in citrate-calcium complexes. For this reason, ionized calcium concentration (both circuit and systemic) and total calcium levels must be calculated, with monitoring advised every 4–8 h and every 12–24 h, respectively.

Hypernatremia

Hypernatremia is seen more commonly in studies using 4% TSC as anticoagulant solution. Due to its high sodium load and inadequate adjustment for tonicity, it

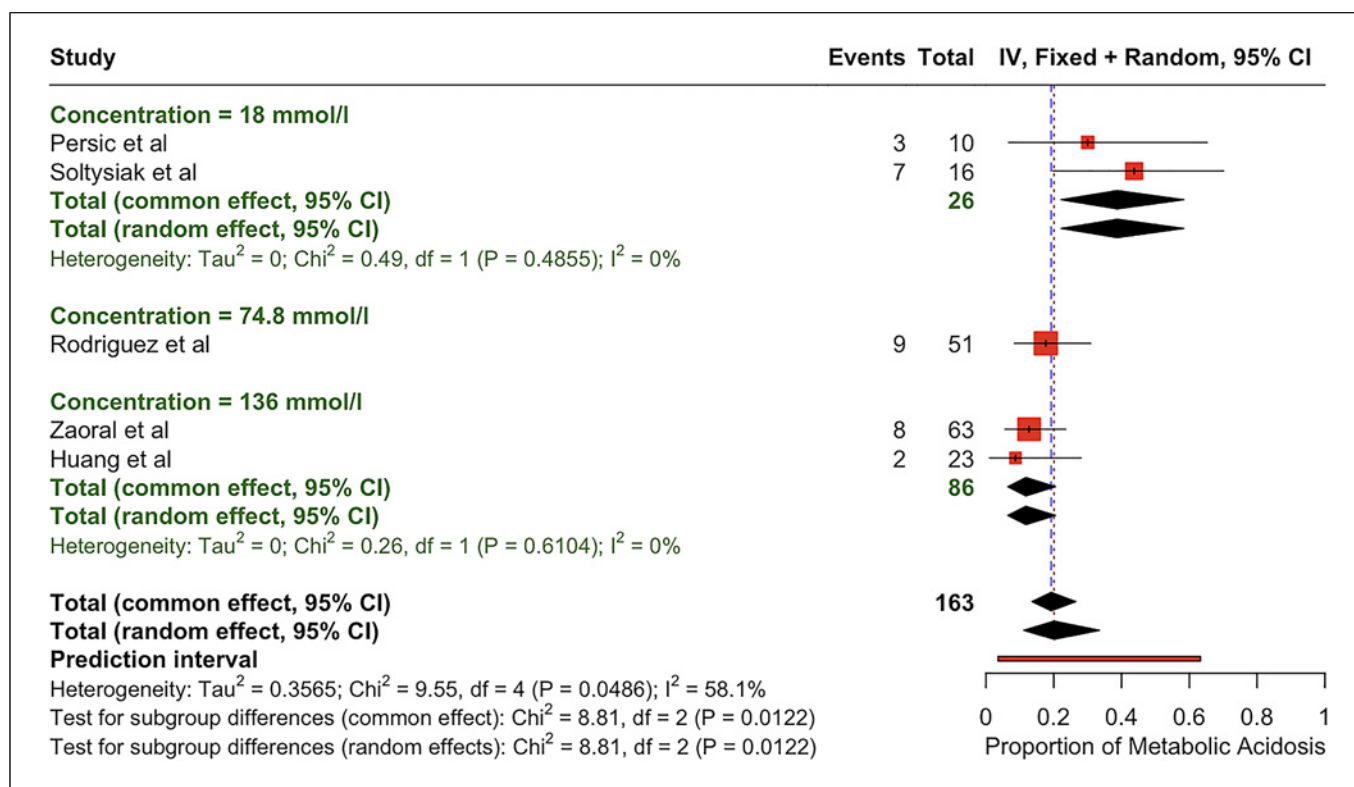


Fig. 3. Adverse events: metabolic acidosis between different RCA concentrations.

can result in hyponatremia [31]. In sum, hyponatremia is a rare complication, which can be easily avoided by adhering to strict RCA protocols and incorporating either low-sodium solutions or replacement fluids [37].

Hypophosphatemia

Persic et al. [13] found the most common electrolyte abnormality to be hypophosphatemia (<1.2 mmol/L). Hypophosphatemia is common, as the continuous therapy avoids the rebound effects and leads a constant phosphate clearance due to the compartmental equilibration [38]. Similarly, Fernández et al. [15] also found that 45.5% of patients developed hypochloremia and 27.3% hypomagnesemia in the citrate group, although this may depend on the citrate formulation. Furthermore, CRRT exhibits high efficiency of phosphate clearance. The most common reason for this metabolic abnormality seems to be a phosphate free dialysate, rather than the citrate solution itself [39].

Hypochloremia

Fernández et al. [15] reported using 18 mmol/L of citrate led to hypochloremia due to low concentration of chloride in prismocitrate solutions, specifically 86 mmol/L of chloride.

Safety of Dilute Citrate vs. Concentrated Citrate

A comparison of results and complications of studies using citrate 18 mmol/L vs. 74.8 mmol/L and 136 mmol/L is shown in Table 2 and Figures 2–6. There was no significant difference between the adverse events among different RCA concentrations.

Our Institute Citrate Protocol for Pediatric CRRT Using Diluted Citrate

We have developed the following citrate protocol for our center and describe our experience with it.

Mode

CRRT using continuous venovenous hemofiltration (using Baxter-Prismaflex) with diluted RCA using an 18 mmol/L concentration of Regiocit/Prismocitrate (given as a pre-blood pump) (components: mmol/L: citrate, 18; sodium, 140; chloride, 86); and post-filter replacement is suggested for pediatric CRRT (Fig. 7). It is important to note that we did not use diffusive component, as it further adds to the dialysis dose, in addition to the prefilter citrate dose.

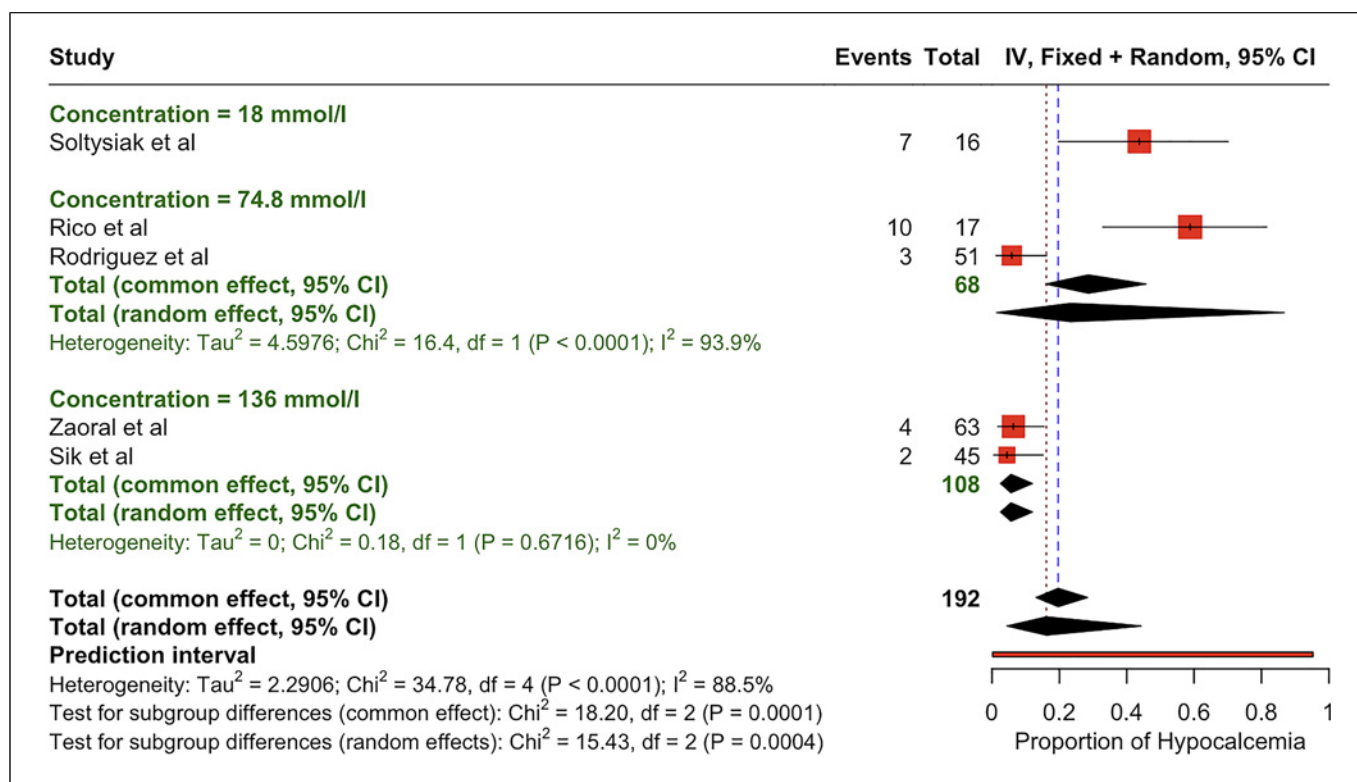


Fig. 4. Adverse events: hypocalcemia between different RCA concentrations.

Prescription

- The CRRT filter and the vascular access selection are to be done according to the patient's weight, with the extracorporeal circuit volume of <10% of the patient's circulatory volume.
- Prismaflex filters are selected as per the patient's size (HF20 for infants with a body surface <0.5 m², Prismaflex M60 set [AN-69 filter; 0.6 m²], M100 set [1 m²] for children weighing >25 kg).
- The usual initial BFR recommended is 3–4 mL/kg per minute up to a maximum of 150 mL/min.
- Prefilter citrate, ordered dialysate, and replacement fluids are used, ensuring appropriate composition and concentration.
- To reduce the risk of hypophosphatemia, phosphate containing post-filter replacement fluid (Biphozyl) (composition after reconstitution, compartments A+B: part A and part B mmol/L sodium 140; 140, potassium 4; 4, magnesium 0.75; 1.5, chloride 122; 122, hydrogen phosphate 1; 2, hydrogen carbonate 22; 22, theoretical osmolarity: 290 mOsm/L, pH: 7.0–8.0) should be considered, which rebalances the patient's metabolic profile.

- The effluent dose should be kept below 35–40 mL/kg/h, unless otherwise specified. Adjust the dialysate and replacement fluid volumes accordingly to achieve the desired fluid balance.
- Set the filtration fraction target to be less than 25%, which may assist in balancing fluid removal and preventing excessive ultrafiltration.

Anticoagulation with Regional Citrate

- Initial citrate dose: 3-mmol/L (Regiocit, citrate concentration 18 mmol/L)
- Initiating citrate-based anticoagulation: For achieving the desired citrate dose, the citrate rate (Regiocit) in cc/h can be calculated as 10 times the BFR of the Prismaflex machine in ccs/min so that if the BFR is 100 mL/min, citrate should be started at 1,000 mL/h, and then the citrate dose is titrated. Calcium compensation with 10% calcium chloride or 10% calcium gluconate should be administered depending on the patient's initial ionized calcium (iCa^{2+}).
- Titrations: An attempt should be made to reduce the citrate flow rate and dose, titrating it with post-filter ionized calcium keeping the levels 0.25–0.35 mmol/L

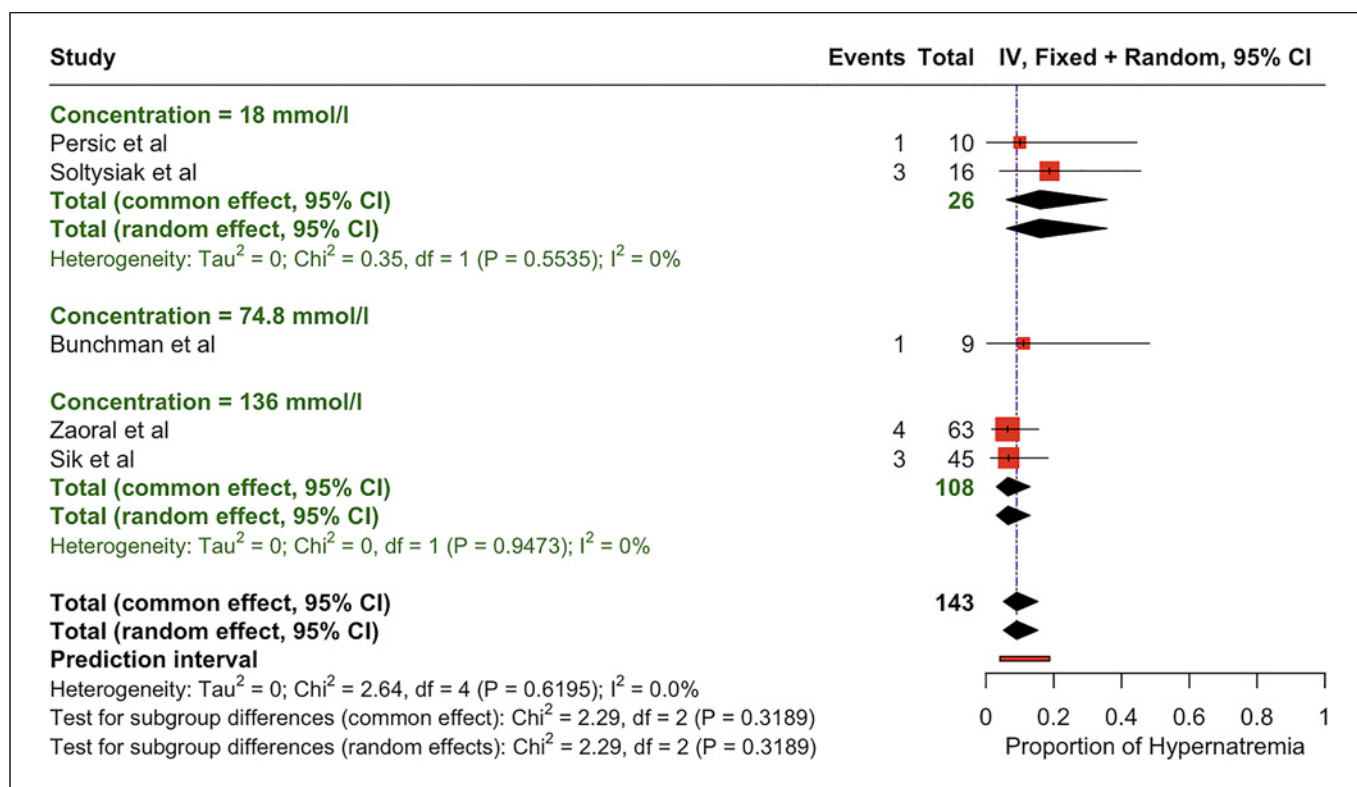


Fig. 5. Adverse events: hypernatremia between different RCA concentrations.

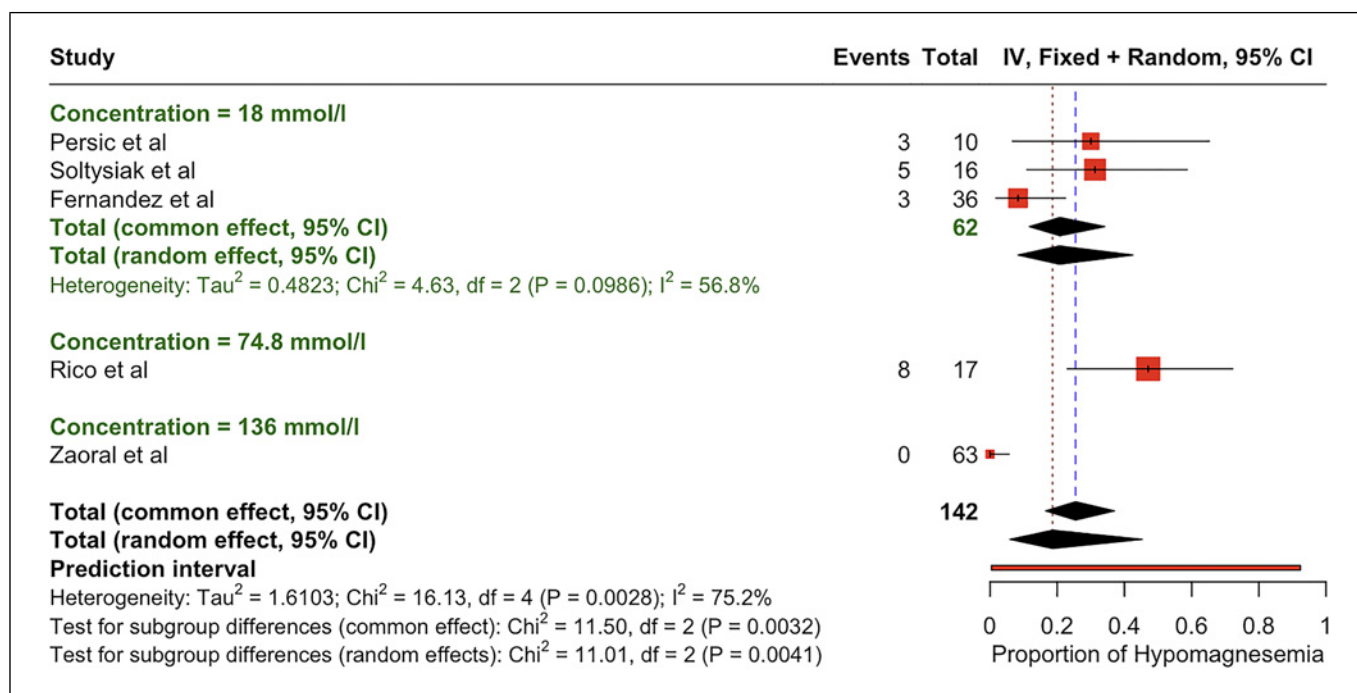


Fig. 6. Adverse events: hypomagnesemia between different RCA concentrations.

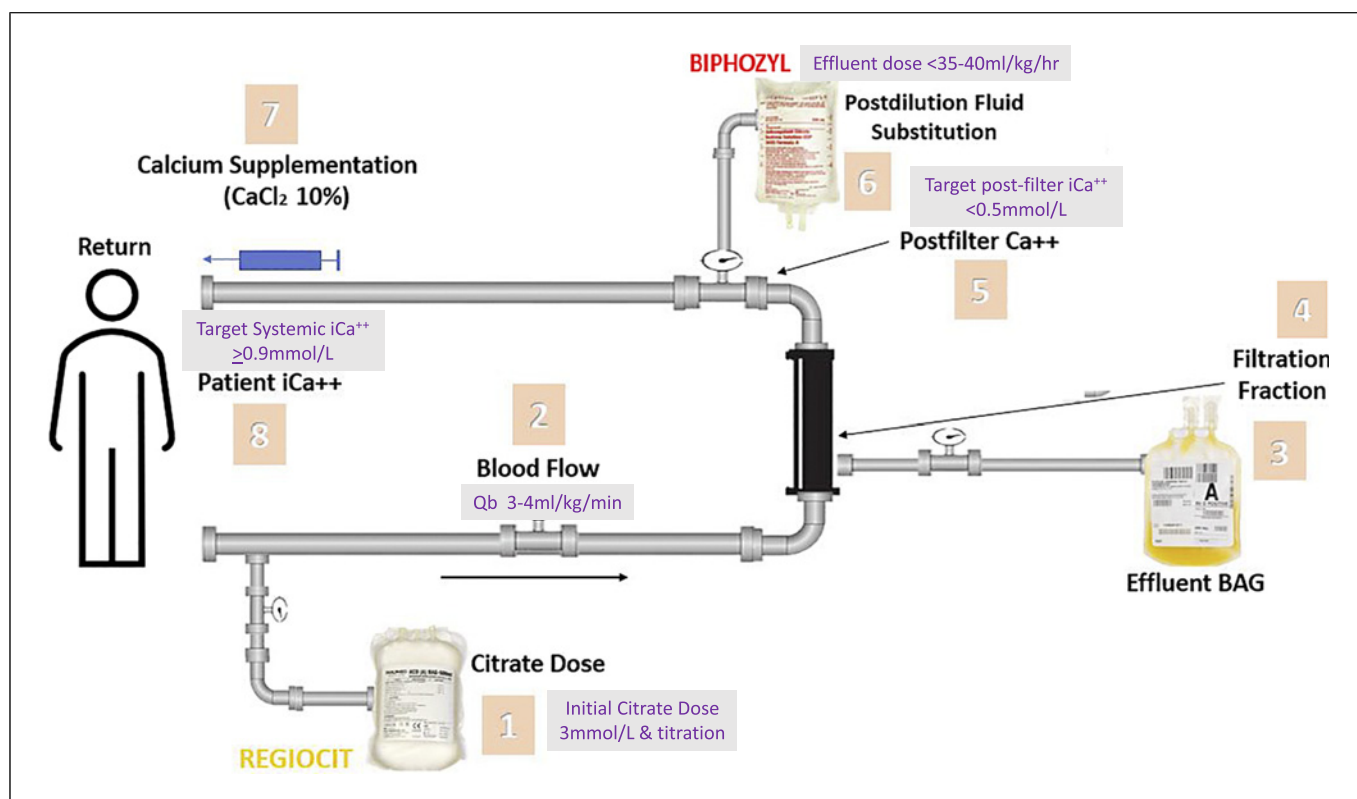


Fig. 7. Diagrammatic representation of CVVH with diluted regional citrate anticoagulation in children with recommended dosage parameters.

and systemic ionized calcium ≥ 0.9 mmol/L. The flow rates in the hemofiltration machine are set as required, considering the patient's fluid status and individual requirements.

Dose Calculations

- Effluent dose rate ($\text{L/h}/1.73 \text{ m}^2$) = $(\text{CFR} + \text{RFR} + \text{NBR}) \times 1.73/\text{BSA} \times 1,000$
 - CFR is the citrate flow rate (mL/h)
 - RFR post-filter replacement fluid flow rate (mL/h)
 - NBR net negative balance rate (mL/h)
 - BSA body surface area (m^2)
- Filtration fraction (%) = $100 \times (\text{RFR} + \text{NBR})/(\text{PFR} + \text{CFR})$
- PFR, i.e., plasmatic flow rate $[(1 - \text{hematocrit}/100) \times \text{BFR} (\text{mL/h})]$.

Treatment Monitoring

When monitoring and managing pediatric patients undergoing CRRT, several critical considerations must be addressed. Prompt attention to low-ionized calcium values is essential due to the adverse effects of hypo-

calcemia. If the patient's ionized calcium falls below 0.9 mmol/L, we administer 10 mL of 10% calcium chloride through a central line. Additionally, maintaining a filter iCa^{2+} concentration of 0.25–0.45 mmol/L is necessary to prevent filter clotting. Regular monitoring of iCa^{2+} is crucial; the patient's iCa^{2+} levels should be checked from the arterial line, while the filter iCa^{2+} should be assessed from the blue port on the Prismaflex machine. A detailed protocol on the titration of citrate and calcium infusions is shown in online supplementary Table 2.

Adjustments to the calcium chloride concentration are made at specific intervals: 30 min and 1 h after initiation of treatment, 30 min after any adjustment until the target ranges are reached, and every 6 h if the calcium concentration remains stable within the target zone. An elevated total calcium to ionized calcium (tCa) ratio (>2.5) is a surrogate marker of citrate accumulation in critically ill patients. It is important to use total calcium and not "corrected calcium" for this calculation. Serum calcium and the tCa ratio are initially checked 6 h after starting treatment and then daily.

Table 3. Clinical characteristics of children undergoing CRRT with diluted citrate at the center

No.	Diagnosis	Other organ system	ICU days	MODS	Age in years	Sex	Inotropes, <i>n</i>	Any complication	Outcomes	Pulmonary edema
1	ARDS, refractory septic shock, severe dengue, AKI	Yes	18	Yes	15	F	2	Nil	Discharged	+
2	Refractory shock, GVHD, ALL, acute pancreatitis, AKI	Yes	25	Yes	9	M	3	Shock, sepsis	Death	+
3	DIC, purpura fulminans, septic shock, severe dengue, AKI	Yes	3	Yes	0.9	M	5	Nil	Discharged	+
4	AML, tumor lysis syndrome, shock, AKI	Yes	6	Yes	0.5	M	1	Shock	Death	+
5	Sepsis, shock, ileal diverticulum, AKI	Yes	7	Yes	6	M	2	MODS, shock	Death	+
6	Refractory shock, acute GVHD, pancreatitis, TMA, hemophagocytic lymphohistiocytosis, ALL, AKI	Yes	4	Yes	9	M	2	Shock	Death	+
7	Severe dengue, multiorgan failure, AKI	Yes	2	Yes	13	M	0	MODS	Discharged	+
8	Aplastic anemia, septic shock, AKI	No	9	Yes	13	F	2	Shock, MODS Cardiac arrest	Death	+
9	Sepsis, shock, ileal diverticulum, AKI	Yes	7	Yes	6	M	2	MODS, shock	Death	+
10	Severe dengue, AKI	Yes	2	Yes	9	M	3	MODS, shock	Death	+
11	Severe dengue, pulmonary hemorrhage, AKI	Yes	3	Yes	16	M	2	MODS, DIC, shock	Death	+
12	Nephrotic syndrome, AKI, seizures	Yes	4	No	13	M	2	None	Discharged	–
13	Severe dengue with severe shock	Yes	5	Yes	16	M	1	MODS	Stable	+
14	Dilated cardiomyopathy	Yes	4	Yes	17	M	2	MODS	Left against advice	+
15	AML, DIC	Yes	4	Yes	12	M	2	DIC	Death	+
16	ITP, septic shock, pseudomonas sepsis, abdominal wall panniculitis, digital gangrene, respiratory failure	Yes	4	Yes	8	M	1	MODS	Discharged	+

AKI, acute kidney injury; ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; DIC, disseminated intravascular coagulation; GVHD, graft vs. host disease; ITP, immune thrombocytopenic purpura; MODS, multiorgan dysfunction syndrome; TMA, thrombotic microangiopathy. ^aAll of these patients had no major complications of CRRT except routine circuit change.

Departmental Audit

Patients

From January 2022 to January 2024, CRRT was performed in 40 children in our pediatric intensive care. Out of these, heparin based or no anticoagulation was performed in

24 children, since they were either <10 kg (*n* = 5) or had liver failure (*n* = 19). Citrate anticoagulation using our structured protocol was used in 16 patients. The diagnosis and clinical profile of all patients are shown in Table 3. All patients were critically ill with multiorgan dysfunction syndrome.

Results

All patients had severe shock, multiorgan dysfunction syndrome, and pulmonary edema ($n = 15$). The mean stay in the ICU was approximately 6.69 days (standard deviation 6.22 days). The mean age of the patients was 10.21 years (standard deviation 5.09 years). The mean number of inotropes used was 2.00 (standard deviation 1.10). Mean BFRs were 4.8 mL/kg/min, and the mean effluent flow rate was 43 mL/kg/h.

The mean duration of CRRT was 49.94 h (standard deviation 18.02 h). The filter life range was 28–104, totaling 799 h. CRRT was interrupted in 8 patients scheduled filter change (4 sessions), circuit clotting (3 sessions), and catheter dysfunction (1 patient). A detailed table showing the CRRT prescription of the patients is shown in online supplementary Table 3.

The citrate dose was reduced (from a baseline initiation dose of 3 mmol/L) in all the patients, after the initiation as all patients received the desired target of post-filter and systemic ionized calcium measurements. The citrate dose was reduced to 2.5 mmol/L in 10 patients. No patient had citrate accumulation. None of the patients had metabolic complications of hypo- or hypernatremia and metabolic alkalosis (range minimum to maximum blood pH 7.01–7.39), with lowest serum phosphorus levels being 3 mg/dL (range 3–5.6). No patient had any bleeding complications.

Conclusions

In pediatric patients treated with CRRT, RCA has been shown to be associated with prolongation of circuit life and decreased incidence of hemofilter change due to clotting, which is in concordance with adult studies. A study from France using a similar protocol reported a median hemofilter life of 24 h (range 7–69 h) only, with the initial lower citrate dose of 2.2 mmol/L [12]. We believe that a citrate dose of 2.5–3 mmol/L is essential to maintain the regional anticoagulation in CRRT. The use of RCA for patients with a significantly greater risk of bleeding or defective coagulation has been advised by the 2012 Kidney Disease: Improving Global Outcomes practice guidelines if there are not any contraindications to citrate [7]. Commercially available solutions, such as Regiocit with an 18 mmol/L citrate concentration, are paired with phosphate-containing dialysate solution, such as Biphosyl, to perform continuous venovenous hemofiltration/CSVHDF. RCA is associated with severe complications especially citrate accumulation, seen especially in vulnerable population such as the pediatric population. Hence, a close monitoring is needed especially in neonates and infants. While our results provide valuable insights on how to

use dilute citrate in sick children on CRRT, they are limited by the sample size and the specific patient population in our pediatric intensive care unit. Given the variations in patient characteristics and clinical outcomes, further multicenter studies with larger cohorts comparing the citrate concentrations are essential to validate our findings. As such, our conclusions are not yet generalizable to all pediatric populations, and additional data are needed to refine treatment protocols and improve patient outcomes. Overall, diluted RCA in CRRT in pediatric patients appears to be feasible, with less complications, and a valuable option for critically ill children. An attempt should be made to lower the CRRT dose, especially in infants, using a higher cutoff of circuit iCa <0.5 mmol/L and a lower threshold of systemic iCa ≥ 0.9 mmol/L.

Statement of Ethics

Ethical approval was taken from the Institutional Ethics Committee. This study protocol was reviewed and approved by Medanta Institutional Ethics Committee (MICR-774/2017). Written informed consent was obtained from their parent/legal guardian/next of kin to participate in the study.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

S.K.S., O.S.R., M.A., A.N., K.S., S.S., and S.B.S.: substantial contributions to the conception and design of the work; the acquisition, analysis, and interpretation of data for the work; and drafting the work and reviewing it critically for important intellectual content. A.T., G.C.B., K.A., T.B., and R.R.: substantial contributions to the conception and design of the work; interpretation of data for the work; and drafting the work or reviewing it critically for important intellectual content. All authors drafted the work and reviewed it critically, approved the final manuscript as submitted, and agreed to be accountable for all aspects of the work. All contributors met the ICJME criteria for authorship.

Data Availability Statement

The data that support the findings of this study are not publicly available due to their containing information that could compromise the privacy of research participants but are available from the corresponding author upon reasonable request.

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